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資料結構

Data Structure

Lab06

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Lab01-Q1.
Blackjack **Complete the queue, initializeDeck, playerTurn, dealerTurn code and add comments.**
Code

```
#include <iostream>
#include <vector>
#include <string>
#include <algorithm>
#include <random>
#include <ctime>
using namespace std;


struct Player { // 玩家結構體，包含名稱和分數
    string name; // 玩家名稱
    int score;   // 玩家分數
};

class Card { //建立一個 Card 類別，用來表示單張撲克牌
public:
    string suit; // 花色
    string rank; // 點數
    Card() {}
    Card(string s, string r) : suit(s), rank(r) {}
    void display() const {
        cout << rank << " of " << suit << endl;
    }
};

class Deck { // 牌組類別，負責生成和洗牌
private:
    Card cards[52]; // 52 張牌
    int index; // 目前發牌索引
public:
    Deck() : index(0) { // 初始化牌組
        string suits[] = { "Hearts", "Diamonds", "Clubs", "Spades" }; // 四種花
色
```

```
        string ranks[] = { "2", "3", "4", "5", "6", "7", "8", "9", "10", "J",
"Q", "K", "A" }; // 13 張牌
        int k = 0;
        for (int i = 0; i < 4; i++) {  // 4 種花色
            for (int j = 0; j < 13; j++) {  // 13 種點數
                cards[k] = Card(suits[i], ranks[j]);
                k++;
            }
        }
    }

    void shuffleDeck() { // 洗牌
        srand(time(0));
        for (int i = 51; i > 0; i--) { // 從最後一張牌開始，向前遍歷
            int j = rand() % (i + 1); // 生成隨機索引 j
            swap(cards[i], cards[j]); // 交換牌組中的兩張牌
        }
    }

    Card getCard(int i) const { // 取得指定位置的牌
        return cards[i];
    }
};

class Queue { // Queue 類別
private:
    vector<Card> deck; // 用來存放卡牌的容器
    int Front;         // 指向牌堆的前端
    int Rear;          // 指向牌堆的尾端
    int capacity;      // 容器最大容量
public:
    // 建構函式
    Queue(int size) : Front(-1), Rear(-1), capacity(size) {}

    // 獲取佇列最前端元素
    Card front(){
        if (isEmpty()) { // 檢查牌堆是否空
            cout << "牌堆為空，無法獲取最前端的牌！" << endl;
```

```
            return Card("", "");
        }
        return deck[Front]; // 回傳最前端卡牌
    }

    // 插入元素到佇列尾端
    void enqueue(Card card) {
        if (isFull()) { // 檢查牌堆是否滿
            cout << "牌堆已滿，無法插入！" << endl;
            return;
        }
        deck.push_back(card);  // 插入元素到尾端
        Rear++;                // 尾端指標移動
        if (Front == -1) Front = 0; // 第一次加入卡牌更改 Front 為 0
    }

    // 從佇列中移除最前端元素
    void dequeue() {
        if (isEmpty()) { // 檢查牌堆是否空
            cout << "牌堆為空，無法發牌!" << endl;
        }
        else {
            Front++; // Front 指標移動
        }
    }

    // 判斷佇列是否為空
    bool isEmpty() const {
        if (Front == -1 || Front > Rear) { // 兩種情況: 牌堆還沒加入牌 or 所有牌都
被 dequeue
            return true;
        }
        else {
            return false;
        }
    }

    // 判斷佇列是否已滿
```

```
    bool isFull() const {
        if (Rear == capacity - 1) { // Rear 在最後一隻張牌的位置
            return true;
        }
        else {
            return false;
        }
    }
};

void initializeDeck(Queue& cardDeck) { // 初始化並洗牌
    Deck deck;
    cout << "初始牌堆: \n";

    for (int i = 0; i < 52; i++) {
        deck.getCard(i).display(); // 印出原始牌堆
    }

    deck.shuffleDeck(); //洗牌
    cout << "\n洗牌後的牌堆: \n";
    for (int i = 0; i < 52; i++) {
        deck.getCard(i).display(); // 印出洗牌後的牌堆
    }

    for (int i = 0; i < 52; i++) {
        cardDeck.enqueue(deck.getCard(i)); // 將洗好的牌堆加入到佇列
    }
}

void initializePlayer(Player* player, string name, Queue& cardDeck) { // 初始化
玩家，發兩張牌
    player->name = name; // 玩家名稱
    player->score = 0; // 玩家分數
    cout << player->name << " 抽到的牌: ";
    Card drawnCard; // 取出的牌

    for (int i = 0; i < 2; i++) { // 發兩張牌
        drawnCard = cardDeck.front(); // 用front() 抽牌並回傳給drawnCard
```

```
        cout << drawnCard.rank << " of " << drawnCard.suit << "  "; // 將抽到的牌印出
        cardDeck.dequeue(); // 從牌堆中移除這張牌

        if (drawnCard.rank == "A") player->score += 1; //計算 A 分數
        else if (drawnCard.rank == "J" || drawnCard.rank == "Q" || drawnCard.rank == "K") player->score += 10; //計算 JQK 分數
        else player->score += stoi(drawnCard.rank); //計算 2-10 分數
    }
    cout << "\n";
    cout << player->name << " 玩家的初始分數: " << player->score << endl; //印出玩家的初始分數
}

void playerTurn(Player* player, Queue& cardDeck) { // 玩家回合
    char choice;
    Card newCard; // 抽到的牌

    while (player->score < 21 && !cardDeck.isEmpty()) { // 玩家小於 21 點且牌堆不為空
        cout << player->name << " 您的手牌分數目前為: " << player->score << " 要抽牌嗎？(h = 抽, s = 停) : "; // 顯示選擇
        cin >> choice; // 輸入選擇

        if (choice == 'h') { // 如果玩家選擇抽牌
            newCard = cardDeck.front(); // 用 front() 抽牌並回傳給 newCard
            cardDeck.dequeue(); // 從牌堆中移除這張牌
            if (newCard.rank == "A") player->score += 1; // 計算 A 點數
            else if (newCard.rank == "J" || newCard.rank == "Q" || newCard.rank == "K") player->score += 10;  // 計算 JQK 點數
            else player->score += stoi(newCard.rank); // 計算 2-10 點數

            // 印出抽到的牌和總點數
            cout << player->name << " 抽到: " << newCard.rank << " of " << newCard.suit << " 總分: " << player->score << endl;

            if (player->score > 21) { //如果玩家大於 21 點
```

```
                cout << player->name << " 爆掉了！遊戲結束。\n"; //玩家爆牌，莊家獲勝
                return;
            }
        }
        else if (choice == 's') { // 如果玩家選擇停牌
            cout << player->name << " 選擇停牌，總分: " << player->score << endl; // 印出總點數
            break;
        }
        else {
            cout << "請輸入有效選項 (h = 抽, s = 停)！" << endl; // 輸入無效
        }
    }
}

void dealerTurn(Player* dealer, Queue& cardDeck) { // 莊家回合
    Card newCard; // 抽到的牌

    while (dealer->score < 17 && !cardDeck.isEmpty()) { // 莊家小於 17 點且牌堆不為空
        newCard = cardDeck.front(); // 用 front() 抽牌並回傳給 newCard
        cardDeck.dequeue(); // 從牌堆中移除這張牌
        if (newCard.rank == "A") dealer->score += 1; // 計算 A 點數
        else if (newCard.rank == "J" || newCard.rank == "Q" || newCard.rank == "K") dealer->score += 10; // 計算 JQK 點數
        else dealer->score += stoi(newCard.rank); // 計算 2-10 點數

         //印出抽到的牌和總點數
        cout << "莊家抽到: " << newCard.rank << " of " << newCard.suit << " 莊家目前總分: " << dealer->score << endl;

        if (dealer->score > 21) { // 如果莊家大於 21 點
            cout << "莊家爆了！玩家獲勝！" << endl; // 玩家獲勝
            return;
        }
    }
}
```

```
void determineWinner(Player* player, Player* dealer) { // 判斷勝負
    if (player->score > 21) cout << player->name << " 爆了！莊家獲勝！\n"; // 玩家爆牌，莊家獲勝
    else if (dealer->score > 21 || player->score > dealer->score)
        cout << player->name << " 贏了！\n"; // 莊家爆牌或玩家點數大於莊家，玩家獲勝
    else if (player->score == dealer->score) cout << "平手！\n"; // 莊家與玩家點數相同，平手
    else cout << "莊家贏了！\n";
}


int main() {
    srand(time(0));
    Queue cardDeck(52); //宣告牌堆
    initializeDeck(cardDeck); //初始化牌堆

    Player player, dealer; //宣告並初始化莊家以及玩家
    initializePlayer(&player, "玩家", cardDeck);
    initializePlayer(&dealer, "莊家", cardDeck);

    playerTurn(&player, cardDeck);
    if (player.score <= 21) {
        cout << "\n 莊家回合...\n";
        dealerTurn(&dealer, cardDeck);
        determineWinner(&player, &dealer);
    }

    return 0;
}
```

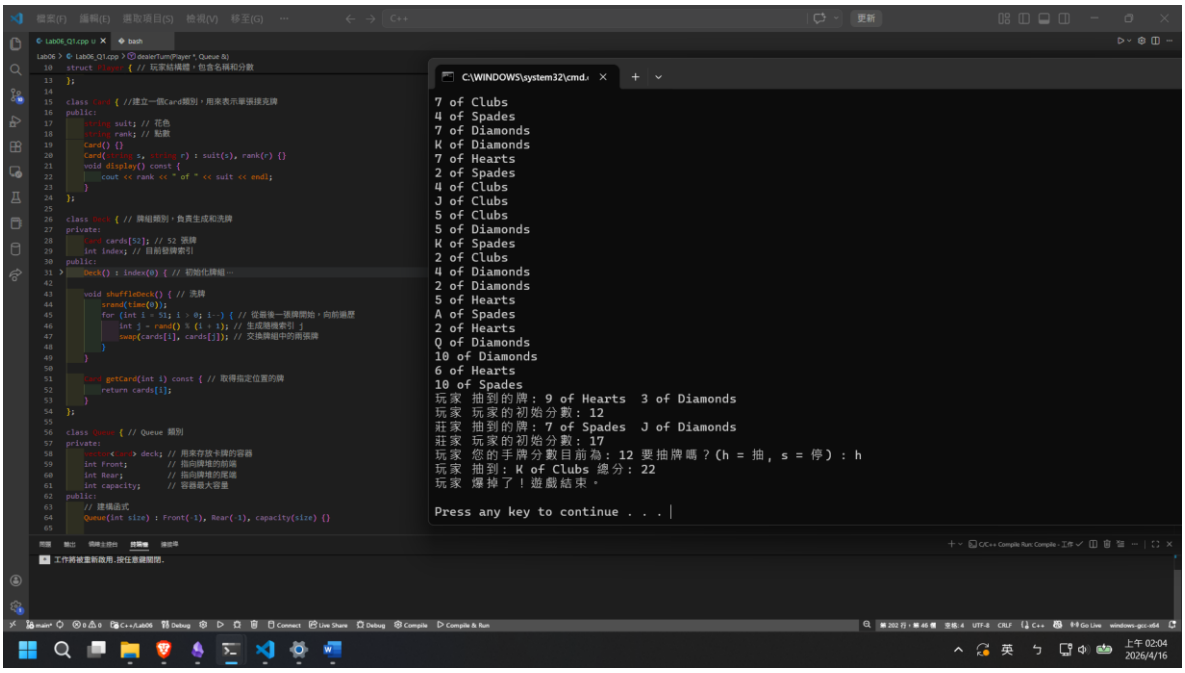
Result

```
初始牌堆:
2 of Hearts
3 of Hearts
4 of Hearts
5 of Hearts
6 of Hearts
7 of Hearts
8 of Hearts
9 of Hearts
10 of Hearts
J of Hearts
Q of Hearts
K of Hearts
A of Hearts
2 of Diamonds
3 of Diamonds
4 of Diamonds
5 of Diamonds
6 of Diamonds
7 of Diamonds
8 of Diamonds
9 of Diamonds
10 of Diamonds
J of Diamonds
Q of Diamonds
K of Diamonds
A of Diamonds
2 of Clubs
3 of Clubs
4 of Clubs
5 of Clubs
6 of Clubs
7 of Clubs
8 of Clubs
9 of Clubs
10 of Clubs
J of Clubs
Q of Clubs
K of Clubs
A of Clubs
2 of Spades
```

```
洗牌後的牌堆:
9 of Hearts
3 of Diamonds
7 of Spades
J of Diamonds
K of Clubs
9 of Spades
4 of Hearts
J of Spades
8 of Diamonds
3 of Spades
3 of Clubs
8 of Spades
A of Diamonds
Q of Clubs
6 of Diamonds
A of Clubs
Q of Hearts
3 of Hearts
6 of Spades
J of Hearts
8 of Hearts
9 of Clubs
5 of Spades
6 of Clubs
K of Hearts
10 of Clubs
8 of Clubs
A of Hearts
9 of Diamonds
10 of Hearts
Q of Spades
7 of Clubs
4 of Spades
7 of Diamonds
K of Diamonds
7 of Hearts
2 of Spades
4 of Clubs
J of Clubs
```

```
A of Hearts
9 of Diamonds
10 of Hearts
Q of Spades
7 of Clubs
4 of Spades
7 of Diamonds
K of Diamonds
7 of Hearts
2 of Spades
4 of Clubs
J of Clubs
5 of Clubs
5 of Diamonds
K of Spades
2 of Clubs
4 of Diamonds
2 of Diamonds
5 of Hearts
A of Spades
2 of Hearts
Q of Diamonds
10 of Diamonds
6 of Hearts
10 of Spades
玩家 抽到的牌: 9 of Hearts  3 of Diamonds
玩家 玩家的初始分數: 12
莊家 抽到的牌: 7 of Spades  J of Diamonds
莊家 玩家的初始分數: 17
玩家 您的手牌分數目前為: 12 要抽牌嗎?(h = 抽, s = 停) : h
玩家 抽到: K of Clubs 總分: 22
玩家 爆掉了!遊戲結束。

Press any key to continue . . .
```



Discussion

Queue: FIFO, Stack: LIFO

Queue 在發牌時從牌的底下開始發牌，第一張放進去的牌，會是第一張發出來的。

Stack 則是從牌的頂端開始發牌，最後一張疊上去的牌，會是第一張被發出來的。